

Non-collisional orogeny in Northeast Japan driven by nearby same-dip double subduction

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Abstract

Same-dip double subduction systems influence plate kinematics, geometry, and mantle flow within the region bounded by the two subduction zones. However, whether these effects extend to nearby plate margin tectonics remains an open question. Furthermore, the range of geodynamic processes operating in these margins and driving non-collisional orogeny (that is, the formation of Andean-type mountain ranges) is not yet fully understood. Here, we explore the potential geodynamic connection between the tectonic evolution of Northeast Japan and the development of the Ryukyu/Izu-Bonin-Marianas same-dip double subduction using three-dimensional geodynamic models. We find that this same-dip double subduction drags the Pacific trench westward, resulting in northward-propagating trench advance and compression affecting the Northeastern Japan arc and back-arc. We thus propose that the dynamics of the Ryukyu/Izu-Bonin-Marianas same-dip double subduction over the past ~ 10 –5 million years (Myr) drove the enigmatic plate kinematics responsible for non-collisional orogeny and back-arc subduction initiation in Northeast Japan since ~ 6 –3.5 Myr

ago, which has made this region prone to catastrophic earthquakes. We also suggest that same-dip double subduction explains various ancient episodes of widespread non-collisional orogenesis and represents a mechanism through which subduction zones establish the plate kinematic conditions necessary for non-collisional orogenesis.

Subduction zones are critical features of our planet, providing the forces needed to move tectonic plates and deform margins, develop volcanic arcs and sedimentary basins, and unleashing the most devastating megathrust earthquakes[1]. In recent years, researchers have observed that these geodynamic settings can be significantly altered by the emergence of a nearby subduction zone, usually forming so-called double subduction systems. These impacts include dramatic changes in plate kinematics, trench motions, slab dips, and the associated mantle flow [2, 3, 4, 5, 6, 7, 8, 9].

Double subduction zones occur in three discernible geometries: divergent double subduction[10], evident in active settings such as the Molucca Sea subduction zones[9] and the Mediterranean region [4]; convergent double subduction, typified by areas hosting oppositely dipping subduction zones, such as those surrounding the Caribbean Plate[11] and Southeast Asia[12]; and same-dip double subduction (SDDS), which feature both ancient and active examples like the Mesozoic Neothetian arc system[6, 3] and the late Cenozoic Ryukyu/Izu-Bonin-Marianas SDDS[13, 5, 2](Fig. 1a). Notably, recent investigations have highlighted the significant geodynamic impact of SDDS initiation, elucidating its role in driving major changes in direction and/or plate velocity [6, 3, 14]. Studies focused on the Ryukyu/Izu-Bonin-Marianas SDDS, involving the subduction of the Pacific Plate beneath the Philippine Sea Plate, which in turn subducts beneath the Eurasian Plate along the Ryukyu-Nankai Trench, indicate that this system leads to Pacific slab steepening and trench advance [13, 5, 2, 15] (Fig. 1a-c). Taken together, these findings suggest that the geodynamics of many subduction zones is best understood when considering these plate boundaries as components

within larger-scale subduction networks. This opens up exciting avenues for understanding active margin dynamics and, more specifically, the magnitudes and length scales at which double subduction is significant. In this regard, while most studies on SDDS have focused on the effects confined to the area of double plate convergence, the broader influence of SDDS on nearby plate margin tectonics remains unknown.

Here, we assess this issue by analyzing the geological evolution of Northeast Japan and the neighboring Ryukyu/Izu-Bonin-Marianas SDDS, presenting evidence for what we define as double subduction-induced orogeny. We hypothesize that the initiation of the Ryukyu/Izu-Bonin-Marianas SDDS modifies trench kinematics along the Pacific subduction, triggering non-collisional compression in Northeast Japan and inducing the active subduction initiation process in the eastern Japan Sea. We test this hypothesis through geologically constrained 3-D geodynamic numerical experiments on double subduction, illustrating how SDDS impacts tectonic stress within the overriding plates of adjacent active margins. The tectonic consequences of SDDS onset illustrate a mechanism whereby complex plate tectonic interactions create optimal plate kinematic conditions for non-collisional orogeny[16].

Late Cenozoic tectonic evolution of Northeast Japan

During the Oligocene-Miocene (~ 25 -20 Myr ago) several marginal basins, including the Japan Sea back-arc basin, and the Shikoku and Parece Vela Basins of the Philippine Sea Plate, underwent simultaneous opening caused by the eastward Pacific trench retreat and major strike slip tectonics[17, 18, 19, 20] (Figs. 1a and 2a,b). This process caused the Japanese Island arc to separate from the Asian continental margin through accelerated back-arc rifting during the Early to Middle Miocene period[20, 17, 21, 22]. This culminated at ~ 16 –15 Myr ago, resulting in the formation of a 3–3.7-kilometer-deep basin with rifted continental crust and local oceanic floor ([e.g., 23, 17, 24, 18])

(Figs. 1a and 2b). In Northeast Japan, neutral stress conditions during thermal sub-
 sidence progressively changed from mild east-west compression since ~ 10 -8 Myr ago
 to highly compressional since ~ 5.6 -3.5 Myr ago [17, 18, 25, 19, 20, 26] (Fig. 2b).
 Pliocene-Quaternary east-west compression led to 10–15 km of shortening within
 Northeast Japan[25], propagating westwards to the Japan Sea[25, 27] and the East
 Korean margin[28]. Currently, the shortening is distributed across several east- and
 west-dipping thrusts and basin inversion structures[25, 27, 20, 19, 26, 29], with rates
 reaching up to 15 mm/yr off Central Japan[30] (Fig. 1b). A compilation map of active
 faulting shows that compression affects a $\sim 1,200$ km-long deformed belt extending into
 the back-arc region west of Hokkaido Island (see Fig. 5 in [20]). The structural obser-
 vations, coupled with the documentation of thrust earthquakes occurring in this area
 over the last century, with magnitudes ranging from Mw 6.9 to 7.8, have prompted
 several authors to propose incipient subduction initiation in the Eastern Japan Sea
 ([e.g., 31, 27, 32]) (Fig. 1a,b). Active back-arc faulting in this region was responsible
 for the recent Noto-Hanto earthquake (2024, Mw 7.6) that generated strong ground
 motion, large crustal deformation and tsunamis that caused significant damage in the
 region[33]. Recent studies also reported geological and geophysical evidence for diffuse
 subduction initiation along the eastern Korean margin active since ~ 5 Myr ago[28]
 (Fig. 1a,b). To the south, tectonics are locally dominated by the interaction of the Izu-
 Bonin arc (Fig. 1a). This area is characterized by a main phase of arc subduction and
 deformation occurring since ~ 6 –5 Myr ago and crustal motion within the active fault
 system controlled by the indented configuration of the plate boundary ([e.g., 20, 34]).

Hypotheses for the Origin of Northeast Japan	185
Compression	186
The origin of the Pliocene-Quaternary compressional tectonic regime across the	187
Northeastern Japanese active margin and the back-arc Japan Sea remains highly con-	188
tentious. One hypothesis suggests a predominant role for the lower plate[35], proposing	189
that the Pacific plate accelerated since ~ 5 Myr ago, introducing a significant com-	190
pression component normal to the Japan trench[36]. However, in this mechanism, it is	191
unclear why compression localized only in Northeast Japan and not along the entire	192
high-convergence western Pacific subduction[37]. Moreover, in global surveys of all	193
major subduction zones, high plate convergence does not correlate with upper plate	194
compression [38]. An alternative, classic hypothesis highlights the impact of eastward	195
upper plate motion, which is attributed to the clockwise rotation of the Amurian plate	196
in response to the distant effects of the India-Eurasia collision [22, 19, 27]. This rotation	197
would have triggered slow convergence between the Amurian Plate and the Okhotsk	198
Plate along the eastern margin of the Japan Sea. Nevertheless, Amurian plate extru-	199
sion has been an ongoing process since at least the late Oligocene-Early Miocene[39],	200
complicating a direct link with the Pliocene onset of compression in northeast Japan	201
and its back-arc region. Furthermore, while the motion of the upper plate towards the	202
trench is significant, it alone does not suffice to initiate compression because upper	203
plate deformation is driven by the relative difference between upper plate and trench	204
motion[16]. It is therefore crucial to examine how the movement of the upper plate	205
interacts with trench dynamics to fully understand the triggering of compression.	206
Both global parametric analyses and simple conceptual/kinematic models indicate	207
that plate margin compression is favored when the trench retreat velocity is less than	208
the upper plate velocity [37]. In the study area, the Japan Trench east of Northeast	209
Japan is advancing at rates of 0.7 to 0.6 cm/yr in a seafloor-spreading aligned absolute	210
reference frame [40], resulting in an overall convergence with the near-stationary upper	211
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231 plate (in the same reference frame), and thus meeting the plate kinematic conditions
 232 for upper plate compression (Fig. 1a). These kinematic conditions are still met even
 233 when analyzing other widely used absolute plate motion reference frames [41, 42, 43].
 234 Based on this, it has been suggested that the presence of a wide Pacific slab, prevent-
 235 ing rapid slab rollback and trench retreat combined with a trenchward-moving upper
 236 plate, created this plate kinematic condition in Northeast Japan[37] (Fig. 1a,b). How-
 237 ever, this conflicts with robust plate reconstructions including mantle slab records
 238 suggesting the subduction of a wide Pacific plate since 50-40 Myr ago ([e.g., 44]),
 239 well before the onset of regional compression in the study area. The Pacific kinematic
 240 conditions driving compression in the study area have also been associated with the
 241 clockwise rotation and northwestward movement of the Philippine Sea Plate[45]. How-
 242 ever, major rotation of this plate occurred since at least ~ 30 -40 Myr ago[44, 46], again
 243 significantly before the onset of compression in Northeast Japan (Fig. 2a,b).

252 Alternatively, recent numerical experiments on Pacific subduction [47] suggest that
 253 the plate kinematics leading to compression may result from a combination of an
 254 imposed eastward asthenospheric flow and slab folding processes at depth, resulting in
 255 several trench retreat-advance cycles since 50 Myr ago. The issue with the applicabil-
 256 ity of this model to Northeast Japan is that these experiments predict an extensional
 257 phase from 6 to 3 Myr ago, whereas geological data indicate strong and continuous
 258 compression since then [17, 18, 25, 19, 20, 26] (Fig. 2b). Moreover, the many alter-
 259 nations of compression and extension since ~ 35 Myr ago, as shown in their modeling
 260 results, do not correlate well with geological data and the overall tectonic evolution of
 261 this specific region (see previous section) (Fig. 2b).

269 Lastly, alternatives such as compression triggered by subduction or accretion
 270 of bathymetric highs, are unlikely due to the localized nature of the associated
 271 compression[37, 38]. More importantly, the absence of a major aseismic ridge or oceanic
 272 plateaus interacting with the northeastern Japan Trench—features necessary to induce
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eastward-directed compression—further diminishes the plausibility of this hypothesis. The only bathymetric high that may influence compression is the Izu-Bonin Arc (Fig. 1a). However, as previously shown[34], it is considered more of a local factor, with its main deformational effects not extending more than approximately 150-200 km north of the accretion zone. Therefore, it cannot account for the documented $\sim 1,200$ km-long region under compression, as evidenced by both active and ancient thrust faults, as well as thrust-related intraplate earthquakes observed over the past few decades (see Fig. 5 in Taira[20]).

In summary, over the past ~ 5 -6 Myr ago, the Northeast Japan arc-to-back-arc region has experienced increasing compression due to the onset of a plate kinematic context associated with an advancing trench. Existing explanations for this phenomenon are either unfeasible[35, 36, 47] or appear incomplete[22, 19, 27, 37, 38, 45]. More importantly, numerical geodynamic models specifically designed to test conceptual frameworks in this region are still lacking.

Double subduction-induced orogeny in Northeast Japan

To investigate the geodynamic mechanisms driving effective plate kinematic conditions for non-collisional orogeny in Northeast Japan over the past ~ 5.6 -3 Myr ago, it is essential to consider trench kinematics and the Cenozoic evolution of Pacific subduction. Present-day trench kinematics[40], spanning from Northeast Japan to the Izu-Bonin-Marianas subduction segments (i.e., Pacific trench in this study), depict a southward increase in westward trench advance (Fig. 1a). This southward shift in trench motion, towards increased advance, is also observed in various trench motion catalogs and is independent of the chosen absolute reference frame [48, 41, 42, 43]. A potential explanation for the induced compression in Japan is then that forces associated with this Izu-Bonin-Marianas trench advance are transmitted northward to this

323 region. In this scenario, the slowly moving upper plate resists the west-directed force
324 associated with trench advance, as transmitted from the south, and that this results
325 in compression in the Northeastern Japan arc and backarc region (Fig. 1a,b). If this
326 is correct, we would expect a temporal correlation between the initiation of trench
327 advance in the Izu-Bonin-Marianas subduction zone and compression in Northeastern
328 Japan and the Japan Sea.

333 The onset of trench advance along the Izu-Bonin-Marianas subduction zone has
334 been linked to a reorganization of major tectonic boundaries associated with the evo-
335 lution of the Ryukyu/Izu-Bonin-Marianas SDDS [13, 5, 2, 15]. Between ~ 40 and 15
336 Myr ago, the Izu-Bonin-Marianas trench predominantly underwent rapid, intermit-
337 tent retreat [17, 18, 19, 20], shifting progressively to trench advance since ~ 10 –5 Myr
338 ago [49, 50, 13, 5, 2, 15] (Fig. 2a,b). Numerical models support the hypothesis that
339 subduction reactivation of the Philippine Sea Plate since 15–10 Myr ago in the Ryukyu-
340 Nankai trench, following a slab detachment caused by ridge subduction [5, 51], exerted
341 a pronounced northwestward pull on the Philippine Sea Plate, consequently driving
342 Pacific trench advance and facilitating the steepening of the Pacific slab (Fig. 2a).
343 Recent numerical studies indicate that the initial stage of Ryukyu subduction (~ 30 –15
344 Myr ago) did not significantly affect the Izu-Bonin-Marianas trench, most likely due
345 to ineffective stress transmission caused by a thermally weakened upper plate at the
346 Shikoku-Parece-Vela Basins backarc spreading center [15]. However, after this system
347 cooled and strengthened/fused since ~ 15 –10 Myr ago, the reactivated Ryukyu sub-
348 duction exerted a strong pull, driving the trench advance observed today [52, 5] (Fig.
349 2b).

361 We note that the significant shift in trench motion for the Izu-Bonin-Marianas
362 trench since ~ 10 –5 Myr ago [49, 5, 50, 52] slightly precedes and coincides with the
363 onset of strong compressional conditions in Northeast Japan and the Japan Sea back-
364 arc basin since ~ 5.6 –3 Myr ago [17, 18, 25, 19, 20, 26, 27, 32, 28] (Fig. 2b). These
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observations indicate a potential causal relationship between the reactivation of the Ryukyu/Izu-Bonin-Marianas SDDS after Philippine Sea plate fusion/strengthening, trench advance, and the onset of the compressional regime in Northeast Japan and the incipient subduction in the Japan Sea.

Geodynamic models of double subduction-induced orogeny

To evaluate the viability of double subduction-induced orogeny in Northeast Japan, we set up 3-D numerical subduction models (see Methods; Fig. 3a-g, Extended Data Figs. 1-6). Our aim was to examine whether the insertion of an additional (Ryukyu-Nankai) subduction zone significantly enhances compression within the Northeast Japan upper plate region relative to a single subduction model containing only Pacific plate subduction along the Japan-Izu-Bonin-Mariana trenches. We therefore modeled two subduction scenarios — with and without subduction along the Ryukyu-Nankai trench (i.e., single vs. double subduction) — using the ASPECT finite element code [53, 54, 55, 56] (Fig. 3a,b).

In our models, we solve for the velocity and stress fields associated with imposed slab and tectonic plate geometries. The plan-view margin shapes are an idealized representation of the present-day plate geometry. The Pacific and Philippine Sea subduction geometries are based on the present-day slabs, as delineated by earthquake hypocenters and seismic tomography [57, 58, 59] (Fig. 1c). We do not model the transition from single to double subduction; rather, we intend to represent the change in stress state from before (~ 10 Myr ago) to significantly after Ryukyu-Nankai subduction reinitiation [5, 51] (present day). We therefore run our “pseudo-instantaneous” single and double subduction models for only 10 kyrs; this is sufficiently short that the plate and slab geometry undergoes minimal evolution from the geometries imposed. In both models, we considered a single, mechanically continuous (i.e., no weak back-arc

area[15]) upper plate, and identical Pacific plate/slab geometries. We adopt a linear viscous rheology for the lithosphere and crust, and we use a simplified power-law viscous rheology for the underlying flowing mantle. This enables us to hone in on the as yet unexplored first-order impact of double subduction on the upper plate stress state, while avoiding complexities associated with a large number of rheological parameters. Here we focus on two models (single and double subduction) with equivalent crustal viscosities ($\eta_{crust} = 5 \times 10^{19}$ Pas) and lithospheric strengths relative to the underlying mantle ($\eta_{lith.}/\eta_{mantle} = 500$). We chose the viscosity values to be intermediate within the ranges inferred from various geophysical, geodynamic, and experimental constraints (e.g., [60, 61, 62, 63, 64, 65]), and to produce tectonic plate velocities compatible with those observed in the Western Pacific, but we have also ran additional tests to verify our first-order conclusions are unchanged if we vary these values (Extended Data Figs. 1, 2).

In the single subduction model, the Pacific trench advances slowly to the left/west at ~ 2 cm/yr and far-field Eurasia is near-stationary (Fig. 3e, Extended Data Fig. 3). Pacific Plate motion is trench-perpendicular and at ~ 8 -10 cm/yr with respect to far-field Eurasia. These modeled velocities exhibit first-order agreement with observed plate velocities in seafloor-spreading aligned [48, 40] or fixed Pacific hotspot reference frames [43]. Adding the Ryukyu-Nankai subduction zone induces a westward Pacific trench advance of 3-4 cm/yr adjacent to the Philippine Sea Plate (i.e., Izu-Bonin-Mariana) (Fig. 3f, Extended Data Figs. 1, 4). This is consistent with observed trench motions [48] and previous subduction modeling [13, 66, 52] (Fig. 1a). These agreements suggest that our models capture the relevant large-scale geodynamic processes and are, in turn, suitable for investigating changes in the large-scale lithospheric stress state induced by double subduction.

Our numerical models show that the establishment of a double subduction system significantly influences the upper plate stress state in the Japan triple junction

and immediately north of this region (Fig. 3e-g and Extended Data Figs. 4, 5). Without double subduction, the lithosphere in the Japan triple junction region is under relatively moderate trench-perpendicular compression of $\sim 10\text{--}20$ MPa (Fig. 3e and Extended Data Fig. 4a). In such geodynamic models, this absolute upper plate stress state is largely controlled by the imposed upper plate size, thickness, density, and strength (e.g., [67, 68]). Here, we therefore focus on the relative change in stress state induced by double subduction, as opposed to absolute stress values. Throughout the upper plate, double subduction induces a significant increase in the magnitude of trench-perpendicular compressive stress (Fig. 3f and Extended Data Fig. 4b). Approximately 350 km onboard of the modeled Japan trench, i.e., in the vicinity of thrusting in Northeast Japan, the average trench-perpendicular compressive normal stress in the lithosphere doubles from ~ 20 MPa to ~ 40 MPa (Fig. 3g and Extended Data Fig. 5). The other normal stress components (vertical and trench-parallel) are only minimally affected by double subduction (Extended Data Fig. 4). This increase in trench-perpendicular compression is due to the along-strike transmission of the Philippine Sea Plate slab pull at the Pacific margin[52, 15], which causes the Pacific slab to indent into the Eurasian upper plate at the Northeast Japan trench, thereby inducing enhanced upper plate compression. The region experiencing increases in horizontal compressive stress extends significantly, covering an area $\sim 1,000\text{--}1,200$ km northward along the Pacific trench and an additional $\sim 1,300\text{--}1,500$ km inland (Fig. 3g). This observation aligns with geological and geophysical evidence of compression, not only within the Japan arc but also in the back-arc domain, which includes a deformed belt approximately $\sim 1,200$ km in length[17, 18, 25, 19, 20, 26, 31, 28, 30, 32, 27] (Figs. 3g and 1a). The model results also highlight subtle compression extending north of northeast Japan, consistent with structural-geomorphological, seismological, and GPS data, all of which indicate mild compression/transpression along Sakhalin Island[69, 18] (Fig. 1a). The observation that the increased compressional lithospheric stress in the

models does not extend beyond approximately $\sim 1,500$ km aligns with newly derived strain maps of eastern Eurasia, which indicate negligible active deformation west of the analyzed area[70] (Figs. 3g and 1a).

Additional tests also demonstrate that, irrespective of the chosen crustal and lithospheric strengths, an increase in trench-perpendicular horizontal compressive stress is a robust feature in transitioning from single to double subduction (Extended Data Figs. 1, 2). The magnitude of the increase in horizontal compression varies with these properties – increasing for higher lithosphere viscosity or lower crustal viscosity – but the single-to-double stress increase is always prevalent. Hence, these results support the double subduction-induced orogeny hypothesis and provide an integrative geodynamic framework that links the evolution of the Ryukyu/Izu-Bonin-Marianas SDDS system [13, 5, 2], the motion of the Philippine Sea Plate, the associated westward drag of the Pacific trench [49, 50, 45], and the onset of arc-backarc compression in Northeast Japan [17, 18, 25, 19, 20, 26, 27, 32, 28]. We suggest that this process accounts for the transition from soft to hard deformation during Izu arc subduction between 8 and 6 Myr ago [20].

Double subduction-induced orogeny in the geological record

Despite the substantial variability in scale and geometrical configuration inherent to highly dynamic double subduction systems, the documentation of active double subduction-induced orogeny in Northeast Japan has significant implications for the orogenic evolution of plate margins proximal to other ancient SDDS systems.

We propose double subduction-induced orogeny, generalized in Fig. 4a, offers a plausible explanation for the widespread Cretaceous compression documented west of the contemporaneous Neotethys subduction system [71, 72] (Fig. 4b). Here, a major change in absolute plate motion direction and the rapid northward drift of the African

plate have been invoked to explain Late Cretaceous compression in Northern Africa [71] (~ 86 – 84 Myr ago, Ayyubid Orogen [73]), subduction initiation along the western Mediterranean region [14], and far-field deformation in Central Europe (~ 84 – 75 Myr, Subhercinian Basin intraplate deformation [74]) (Fig. 4b). Recent studies suggest that the onset of a $\sim 12,000$ km long SDDS at approximately 105 Myr [72], along with the initiation of effective slab pull around ~ 95 Myr, coincided with the directional shift and acceleration in the drift of the African plate [14]. Updated plate kinematic data indicate that Africa–Eurasia relative plate motion along the western Mediterranean active margin transitioned from very slow motion (< 5 mm/yr) in a south-southeast direction between ~ 125 and 90 Myr ago to a northeast direction with a velocity of 30–20 mm/yr from ~ 95 to 70 Myr ago [14] (Fig. 4b). Although differing in dimensions, exact configuration, and partially in the nature of the orogenic processes it involves—such as the collision of Iberia—we propose that, within the framework of the double subduction-induced orogeny concept introduced here (Fig. 4a), a potential genetic connection exists (Fig. 4b). Specifically, this connection links the lifespan (105–75 Myr ago) of the large neighboring SDDS zone in the Neotethys Ocean with the associated plate kinematic change and the plate-wide compression (~ 90 – 75 Myr ago [72, 14]) observed to the west of this system (Fig. 4b).

Although more speculative due to the limited precision of early Paleozoic plate margin reconstructions for southwest Gondwana, another potential candidate for ancient double subduction-induced orogeny is the Paleozoic southern South American margin. Notably, the Chaitenia SDDS (~ 380 – 340 Myr ago) to the west of Patagonia [75] partially overlaps in time with the Chanic orogeny (~ 385 – 350 Myr ago) immediately to the north of this ancient SDDS [76]. If active subduction characterized the entire margin at this time, as recently suggested [77], and there was a north-south mechanical connection between the Chaitenia subduction and that to the north, it is possible that the onset of the Chaitenia SDDS may have contributed to sustaining, if not driving,

599 the protracted Chanic orogeny north of Patagonia. Although this hypothesis provides
600 an elegant explanation for the required trench advance to drive back-arc closure of
601 the Cuyano-*proto-ocean* basin, it warrants further investigation in light of more pre-
602 cise information on the nature and configuration of plate margins during the early
603 Paleozoic.

607 The case of double subduction-induced orogeny in Japan illustrates how subduction
608 interactions, linked to the emergence of a nearby subduction zone, can trigger far-field
609 effects that cascade onto neighboring plate margins. This perspective offers valuable
610 insight into the interpretation of the seismotectonic setting of Northeast Japan, a
611 region renowned for its devastating seismic events, including the Tohoku megathrust
612 earthquake (2011, $M_w = 9.0$) [35] and the recent Noto Peninsula intraplate earthquake
613 (2024, $M_w = 7.6$) [33] in the back-arc area. With this understanding of the tectonic
614 impact of SDDS zones, we anticipate that a more thorough examination of ancient
615 plate margin geological records will uncover additional cases of double subduction
616 zone-induced orogeny.

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Author contributions	645
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G. M. G. conceived the study. Z. G. and A. F. H. designed and carried out the	647
numerical modeling. C. F. provided background on the study area and geodynamic	648
processes. All authors contributed to writing and editing the manuscript, as well as	649
creating the figures.	650
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Competing interests	655
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The authors declare that they have no competing interests.	657
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691 **Figures**

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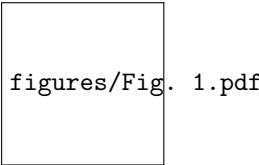


Fig. 1 Present-day plate kinematic setting and mantle structure of the western Pacific subduction zone. **a** Map indicating the major tectonic plates in the study area, subduction isodepths[57], absolute trench and upper plate kinematics [40], the location of Northeast Japan Island and the Japan Sea, and the Ryukyu-Nankai/Izu-Bonin-Marianas SDDS system. This map also shows the location of a composite structural cross-section from the northeast Japan arc[29] to the backarc region up to the eastern Korean margin[28] shown in **b**. **c** Seismic tomography sections[59] showing the mantle structure and slab seismicity associated with the Northeastern Japan subduction and the Ryukyu/Izu-Bonin-Marianas SDDS system. Abbreviations are: EUR: Eurasian Plate; AP: Amurian Plate; PSP: Philippine Sea Plate; PAC: Pacific Plate; OP: Okhotsk Plate; HI: Hokkaido Island; SI: Sakhalin Island; NEJ: Northeastern Japan; SW: Southwest Japan, JS: Japan Sea; R: Ryukyu trench, N: Nankai trench, IB: Izu-Bonin subduction segment; M: Marianas subduction segment; and SVPB: Extinct Shikoku-Parece-Vela backarc basin.

Fig. 2 Cenozoic kinematics of Izu-Bonin-Marianas trench and its relationship to deformation events in the arc and backarc regions of Northeast Japan. **a** Evolution of the Izu-Bonin-Marianas trench migration in the late Cenozoic from Faccenna *et al.* [5] that considers a moving Indian/Atlantic hotspot reference frame[41]. **b** Time correlation of tectonic and kinematic changes in the study area. Please note that the progressive shift in the motion of the Izu-Bonin-Marianas Trench, following PSP fusion and the reactivation of subduction along the Ryukyu-Nankai Trench, slightly precedes and overlaps with the onset of strong compressional conditions in Northeast Japan and the Japan Sea back-arc basin. Abbreviations are: IBM: Izu-Bonin-Marianas Trench, EUR: Eurasian Plate; PSP: Philippine Sea Plate, and PAC: Pacific Plate.

figures/Fig. 3.pdf

Fig. 3 3-D numerical subduction models testing the influence of the Ryukyu/Izu-Bonin-Marianas SDDS on driving upper plate compression in the northeastern arc-to-backarc regions of Japan. Set up for **a** single and **b** double subduction experiments, with the viscosity and velocity fields in cross sections shown in **c** and **d**, and the computed trench-perpendicular (x-direction) normal stress field and plate velocity displayed in **e** and **f** for the reference single and double subduction models, respectively. Stress and velocity is extracted within the lithospheric plates at 30 km depth. **g** Shows the normal stress difference between the models indicating significantly enhanced compression in the area corresponding to the location of Northeast Japan. For simplicity, the Okhotsk Plate is not included, and only an equivalent to the Eurasian Plate is considered. Abbreviations are: EUR: Eurasian Plate; PSP: Philippine Sea Plate, and PAC: Pacific Plate.

figures/Fig. 4.pdf

Fig. 4 Generalized conceptual model of double subduction-induced orogeny and a proposed Mesozoic case associated with the Neotethys SDDS system. **a** A conceptual model illustrates the impact on trench dynamics, transitioning from trench retreat to trench advance, triggered by the onset of a same-dip double subduction system (SDSS) on a plate capable of transmitting slab pull between trenches. This model illustrates the tectonic effects of this process, highlighting plate margin compression occurring away from the SDSS. For simplicity, the upper-right panel depicts a fixed upper plate. The conceptual framework builds upon the established understanding of SDSS dynamics[6, 2, 5, 3, 15, 13, 14] and our own results on the tectonic effects on near by plate margins. **b** Proposal of a double subduction-induced orogenic event based on the documentation of Late Cretaceous plate margin and intraplate compression[74, 73, 71] west of the contemporaneous Neotethys SDDS system[14]. Global and regional plate kinematic reconstructions for the late Early to Late Cretaceous are from Gürer *et al.*[14] and Müller *et al.* [39], respectively.

875 **Methods**

876 877 **Numerical subduction modeling**

879 Our 3-D subduction models were constructed using the ASPECT finite element code
880 (version 2.5.0) [53, 54, 56, 55]. ASPECT was used to solve the conservation of momen-
881 tum (Stokes equation) and mass (continuity equation) for an incompressible viscous
882 fluid (Boussinesq approximation). Our models are purely compositional. That is, we
883 neglect temperature/thermal diffusion and density and viscosity are purely depen-
884 dent on compositional fields. Our models contain four compositional fields that have
885 distinct material properties: the crust, upper plate (Eurasia) lithosphere, subducting
886 plate (Philippine Sea Plate and Pacific Plate) lithosphere, and stiffer material placed
887 at the lateral edges of the Pacific and Eurasian plates (e.g., see Extended Data Fig.
888 3a,c). The background material corresponds to the sub-lithospheric mantle and has
889 another set of mechanical properties (see later paragraphs of this section). The lateral
890 plate edges are imposed to be stiffer (higher viscosity) to retard subduction initiation
891 in these locations. Our models are “pseudo-instantaneous” in that we solve for the
892 velocity and stress fields associated with the imposed slab and plate tectonic geome-
893 tries, as opposed to allowing the structure/slabs to advect for millions of years. In
894 practice, we extract the solution after 10,000 years of model evolution. Given mod-
895 eled plate velocities of ≤ 10 cm/yr, we do not observe noticeable advection of the
896 mantle/tectonic plates over this duration.

899 We use ASPECT’s Adaptive mesh refinement (AMR) functionality to highly refine
900 the finite element mesh within the lithospheric plates – including the crustal layer that
901 makes up the plate boundary shear zone – and within other regions containing large
902 viscosity gradients (e.g., the upper-to-lower mantle transition). In these regions, we
903 have a maximum resolution corresponding to ≈ 4 km wide cubic finite elements. The
904 lowest resolution occurs far away from the tectonic plates and corresponds to ≈ 60 km
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wide cubic finite elements. This approach results in $\sim 5,500,000$ cells unequally distributed throughout the model domains. Simulations were ran on the Skylake (SKX) nodes of the Texas Advanced Computing Center’s Stampede3 cluster.

Our subduction models have a Cartesian domain spanning 8000 km (trench-perpendicular) $\times$ 5000 km (trench-parallel) $\times$ 1000 km (depth). There are no external forces or velocities applied to the subduction system and all model boundaries are mechanically free slip. In our single subduction case, we model Pacific subduction beneath a homogeneous Eurasian plate (Fig. 3a). In the double subduction case, we add the Philippine Sea plate and Ryukyu-Nankai subduction, but leave the Pacific plate/slab geometry completely unchanged (Fig. 3b, Extended Data Fig. 6). At the northern edge of the subduction system ($y = 5000$ km), we impose a “free” plate boundary (i.e., tectonic plates are not attached to the model boundary; Extended Data Fig. 3a,c). This enables the Pacific plate to have a north-south velocity component in this northerly model region. At the other side of the subduction system ($y = 0$), the plates are “fixed” to the model boundary. This boundary acts as a plane of symmetry, thereby enabling us to model the diamond-shaped Philippine Sea plate as a triangle (Fig. 3a,b and Extended Data Fig. 3).

We constructed the input plate and slab geometries using the Geodynamic World Builder tool [78, 79]. We included only the tectonic complexities that are of first-order importance for our proposed mechanism. Our plate boundaries are linear, aside from the protruding cusp of the Pacific trench in northern Japan. This north south segment is 1000 km long and at an angle of 27° relative to the 1600 km-long northern (Kuril-Kamchatka) and southern (Izu-Bonin) Pacific sections (Fig. 1a). In the double subduction case, the Philippine Sea plate was added by including the Ryukyu-Nankai trenches (Fig. 1a). This Philippine Sea Plate-Eurasia plate boundary is 1700 km long and at 29° to the Japan trench. We adopt a constant Pacific slab geometry from north-to-south, with the base of the slab extending to 625 km depth. Of the 1700 km long

967 Philippine Sea Plate-Eurasia plate boundary – in the double subduction system – the
 968 southern 1250 km corresponds to deeper Ryukyu subduction (400 km slab depth) and
 969 the northern 450 km corresponds to shallower Nankai subduction (210 km slab depth)
 970 [44, 57, 58]. Detailed slab geometries are displayed in Extended Data Figure 3 (cross-
 971 sections) and Figure 6 (3-D visualization). Both the Pacific and Philippine Sea Plate
 972 slabs have a slab-top radii of curvature of 185 km and, below this curved portion, dips
 973 of 60 deg. To approximate the interaction of the Pacific slabs with the higher viscosity
 974 lower mantle, we linear reduce the dip to zero as a function of along-slab distance
 975 along the deepest 200 km of the slab. We impose lithospheric thicknesses of 90 km
 976 and 60 km for the older and younger Pacific and Philippine Sea Plates, respectively,
 977 and a 60 km thickness for the idealized Eurasian plate. A weak, 10 km thick crustal
 978 layer is place on top of all plates and along the upper surface of slabs to 70 km depth.
 979 This facilitates plate boundary decoupling, thereby enabling us to reproduce the main
 980 features of observed plate and trench velocities (e.g., Fig. 1a, [48, 40, 43]). To ensure
 981 the Philippine Sea Plate (and Nankai slab) and Eurasian plate are also decoupled, a
 982 ≈ 150 km long and 50 km wide weak zone, with the same properties as the background
 983 mantle, is placed at the northern edge of the Philippine Sea Plate and Nankai slab.
 984 This weak zone extends through the entire plate and has two components: an east-
 985 west (x-directed) component that delineates the northern/top edge of the Philippine
 986 Sea Plate, and a southeast-northwest component that is parallel to the lateral edge of
 987 the northwest dipping Nankai slab (see Extended Data Fig. 6 for a 3-D view of the
 988 weak zone). The latter is needed to stop the Nankai slab from intersecting/coupling
 989 with the overriding Eurasia. To check that the specific geometry of this weak zone does
 990 not impact first order model behavior, we ran an additional double subduction model
 991 with a wider weak region in the triple junction region. In this case, weak material
 992 fills the entire region (triangle) between the northern edge of the Nankai slab and its
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projection onto the Japan Trench. Neither the stress pattern or stress magnitudes are significantly affected (cf. Extended Data Figs. 4b,c).

The Pacific and Philippine Sea lithospheric plates have constant densities of 3350 kg/m³; 50 kg/m³ denser than the sub-lithospheric mantle. The Eurasian plate and crustal layers are neutrally buoyant with respect the mantle, with densities of 3300 kg/m³. This is to approximate the lower densities of continental lithosphere and basaltic crust relative to oceanic lithosphere. The lithospheric and crustal layers are isoviscous and the underlying sub-lithospheric mantle has a composite Newtonian and power-law rheology (e.g., [80]) to mimic the first-order effects of diffusion and dislocation creep, respectively. Lithospheric plates are set to be 500 times more viscous (5×10^{22} Pas) than the reference mantle viscosity (10^{20} Pas). This is within the 10-1000 range of slab-mantle viscosity contrasts generally suggested by geoid constraints [61, 62], slab shapes and trench motions [81, 82, 60], and plate bending considerations [83, 63, 84]. We consider isoviscous plates, i.e., without plastic yielding, to both simplify our model setups and minimize non-linearity (and hence speed up model convergence). While this limits lithospheric deformation to diffuse viscous deformation distributed over 100s-to-1000s of km, as opposed to more "Earth-like" localized shear zones, it enables us to examine which regions experience high viscous stress and hence would ultimately be prone to yielding. We increase the lithosphere viscosity by an order of magnitude within 1000 km wide (trench-perpendicular) edges of the Eurasian and Pacific Plates; this is to ensure subduction does not initiate at the edges. This higher plate edge viscosity causes the stress discontinuities noticeable at $x = 1000$ km and 6000 km (Fig. 3e,f). The crustal layer has a viscosity of 5×10^{19} Pas. Considering a deep plate interface temperature of, e.g., 500-600°C, and strain rate of a 10^{-12} s⁻¹, this viscosity overlaps with that recorded for blueschists, and resides between that of stronger meta-basalts and weaker metasediments and serpentinite [64, 85, 86, 87, 88]. It is also within the 10^{18} - 10^{20} Pas range estimated using numerical shear experiments

for the global range of exhumed shear zones lithological distributions [65]. However, given our simplified shear zone implementation (a single isoviscous layer), the most appropriate effective shear zone viscosity is uncertain, we therefore test a range of values (Extended Data Fig. 1).

Power-law flow in the sub-lithospheric mantle is included due to its large effect on plate and trench motions (e.g., [89, 90]). Due to the compositional nature of our models, we neglect the temperature dependence of mantle viscosity, and additionally omit the pressure dependence. The Newtonian viscosity component of the mantle, η_N , is set to 2×10^{20} Pas. The power-law component, η_P , is calculated following:

$$\eta_P = \eta_N (\dot{\epsilon} / \dot{\epsilon}_T)^{\frac{1-n}{n}} \quad (1)$$

where $\dot{\epsilon}$ is the square root of the second invariant of the strain rate tensor, n is the power-law exponent ($= 3.5$), and $\dot{\epsilon}_T$ is the transition strain rate. Following Holt and Becker[90], we set $\dot{\epsilon}_T = 5 \times 10^{-15} \text{ s}^{-1}$. For strain rates at this value, $\eta_P = \eta_N = 2 \times 10^{20}$ Pas. The effective model viscosity is then calculated as:

$$\eta_{eff} = \left(\frac{1}{\eta_N} + \frac{1}{\eta_P} \right)^{-1} \quad (2)$$

which produces the reference mantle viscosity of $\eta_{eff} = 10^{20}$ Pas at $\dot{\epsilon} = 5 \times 10^{-15} \text{ s}^{-1}$. Given that significant uncertainty remains regarding the appropriate creep law parameters (e.g., [91]), even for the dominant upper mantle mineral olivine, we consider our simplified mantle rheology to be appropriately complex. Beyond 660 km depth, the lower mantle is set to a viscosity of 5×10^{21} Pas (i.e., a factor 50 increase relative to the reference upper mantle viscosity). This follows geoid and slab sinking rate studies that indicate the lower mantle has a higher viscosity than the upper mantle (e.g., [92, 93]).

To verify the first-order behavior of our models does not depend on the assigned crustal and lithospheric viscosities, we have ran additional models with different values. Relative to the reference crustal viscosity (5×10^{19} Pas), we test a lower value of 10^{19} Pas and higher values of 2.5×10^{20} Pas and 10^{20} Pas (Extended Data Fig. 2). Relative to the reference slab-mantle viscosity ratio (500), we test higher and lower values of 2000 and 125 (Extended Data Fig. 3). For each of these additional parameter tests, we turn off power-law flow (i.e., Equation 1) within the sub-lithospheric mantle and instead use a constant (Newtonian) mantle viscosity of 10^{20} Pas. Removing this complexity does not substantially affect the modeled stress fields (cf. Fig. 3e-g and Extended Data Figs. 2 and 3), but reduces model run times to $\approx 30\%$ of the equivalent power-law models. Overall, increasing the slab-mantle viscosity ratio and/or decreasing the crustal viscosity increase the magnitude of the double subduction induced stress increase. However, there is always an increase in horizontal compression due to double subduction.

Data availability

All data needed to evaluate the conclusions in the paper are presented in this manuscript, the Extended Data figures, and the online repository containing the model input files. Model input geometries (e.g, plate and slab shapes) were constructed using the Geodynamic World Builder, also available on GitHub: <https://github.com/GeodynamicWorldBuilder/WorldBuilder>. We provide all ASPECT and Geodynamic World Builder input files needed to run the models as a data publication on Zenodo (DOI: <https://doi.org/10.5281/zenodo.15090347>, Gianni *et al.* [94]).

Code availability

All models were run using ASPECT version 2.5.0, which is freely available on GitHub: <https://github.com/geodynamics/aspect>.

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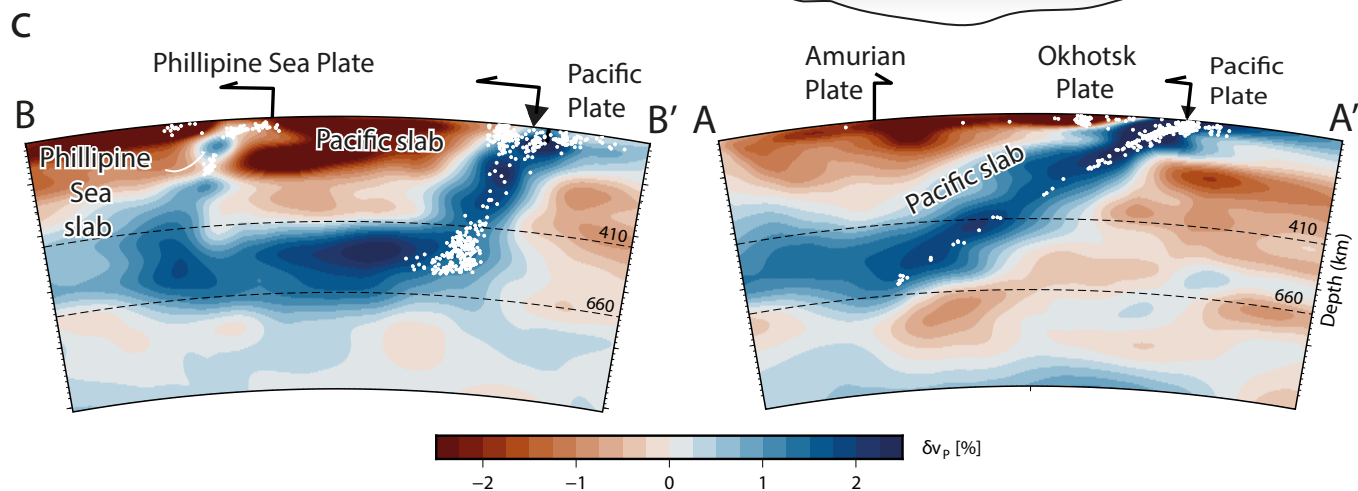
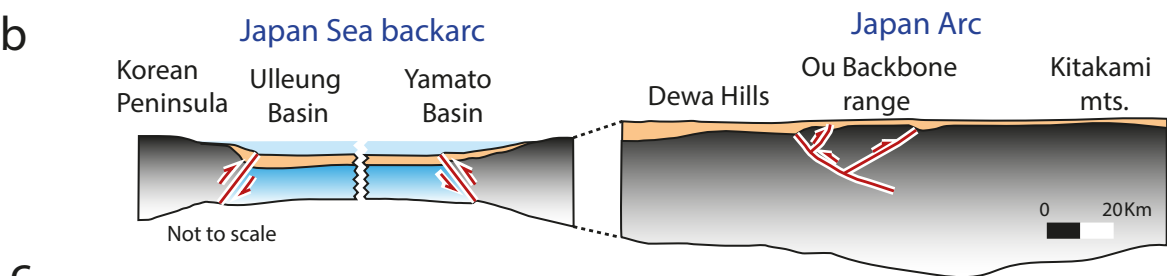
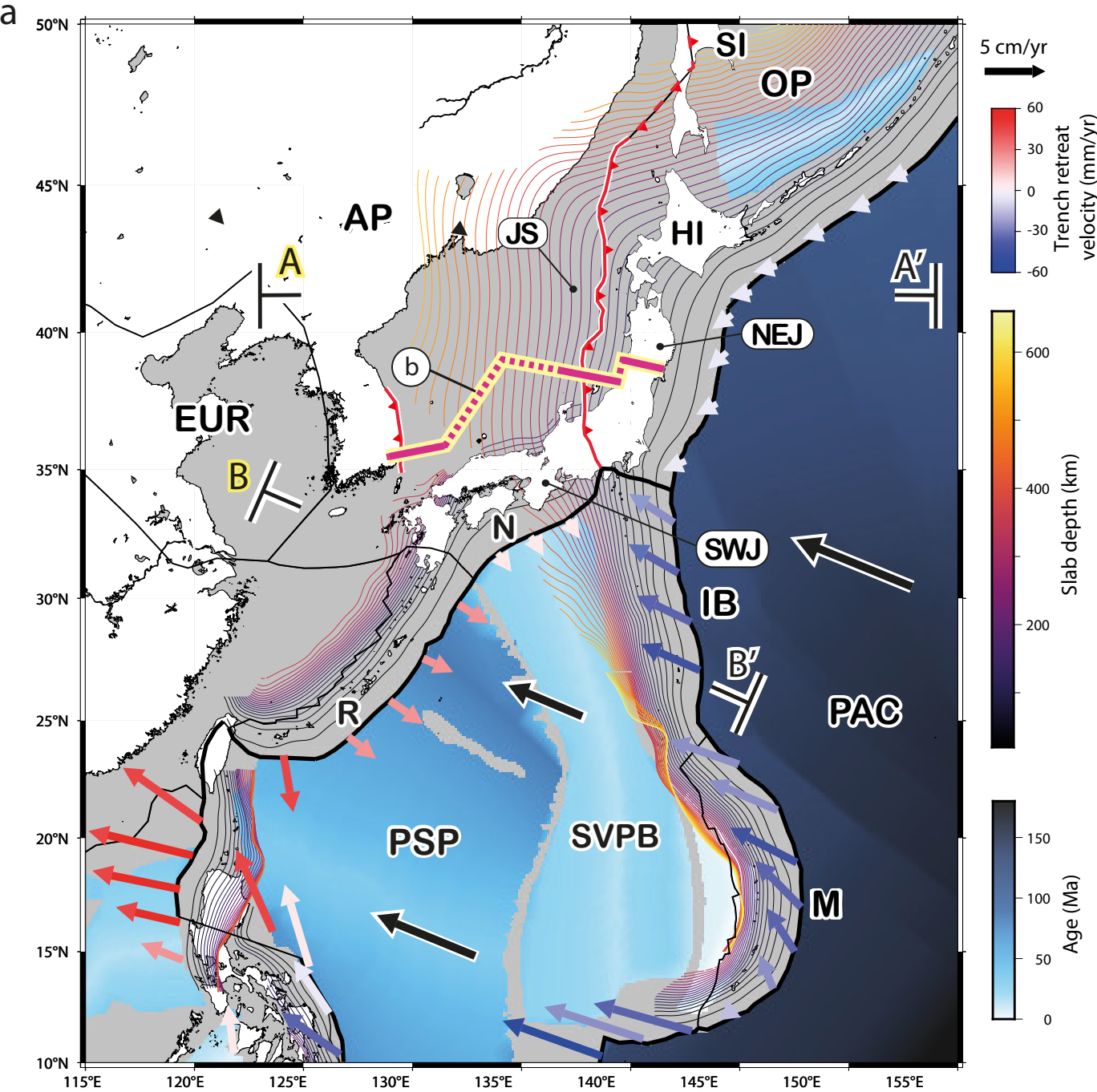
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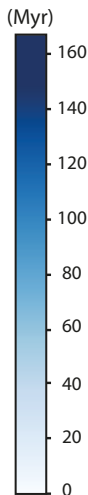
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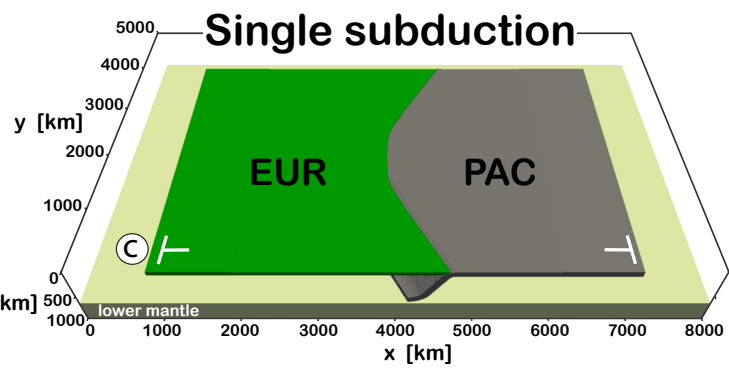
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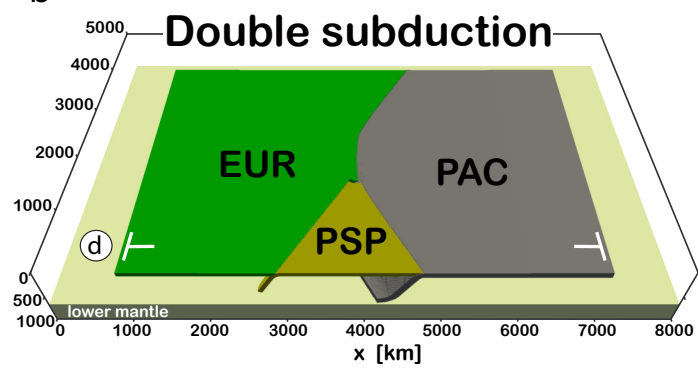


b

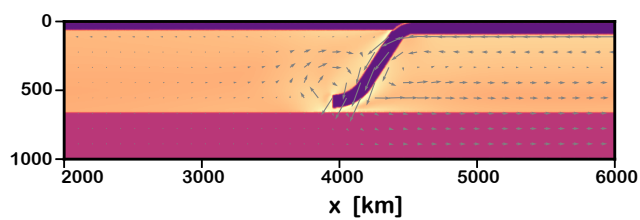
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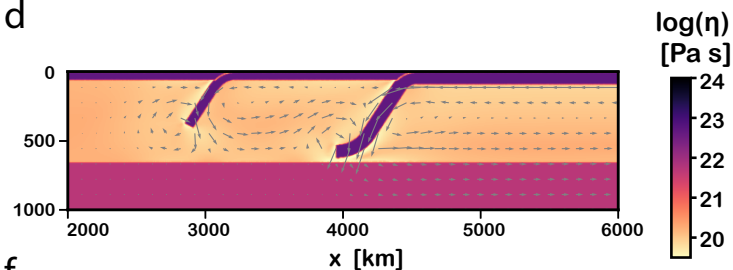
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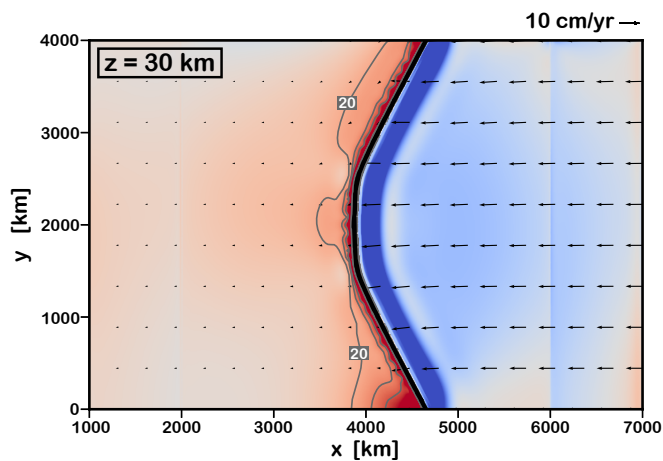
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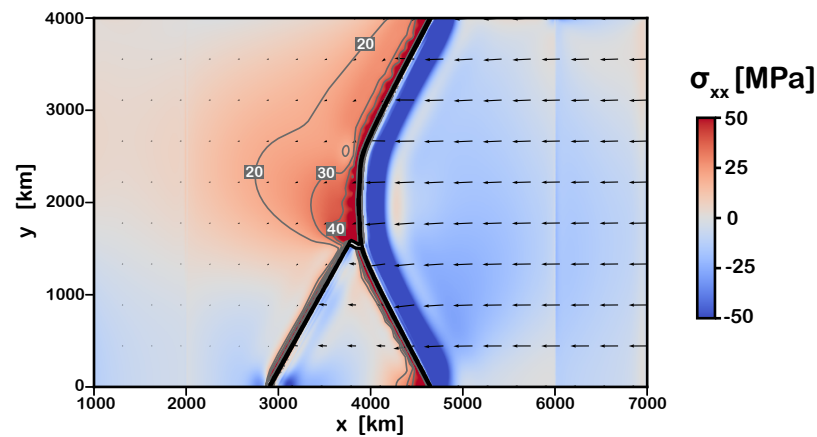
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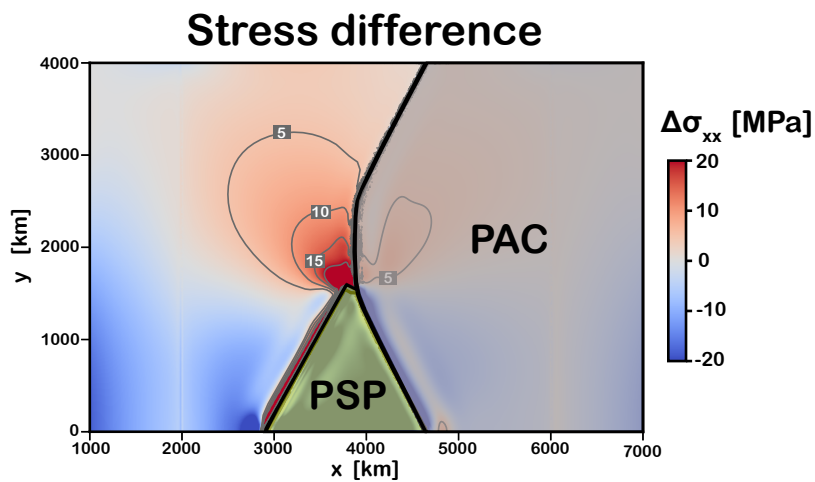
e



f



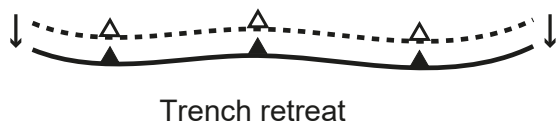
g



a

Single subduction

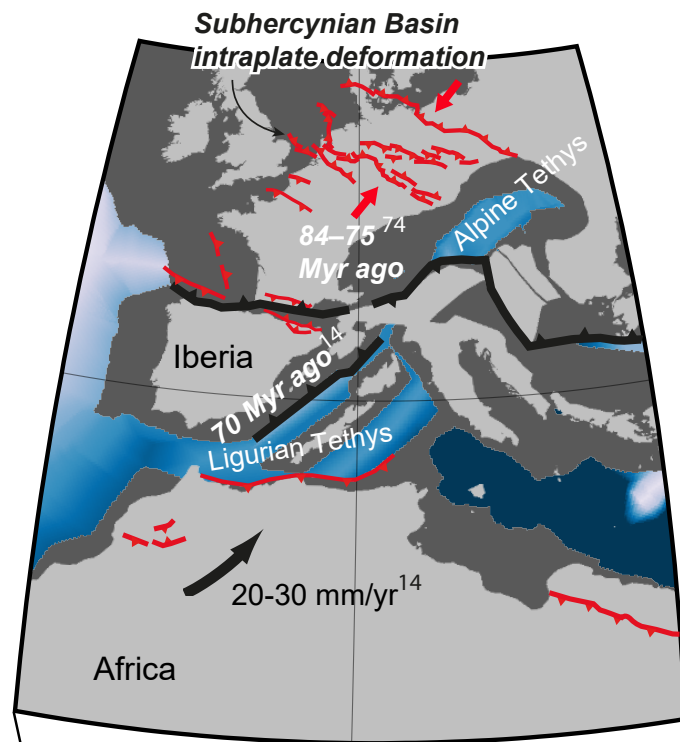
Active backarc
mid-ocean ridge (MOR)



SDDS system

MOR extinction
/fused

Upper plate
compression

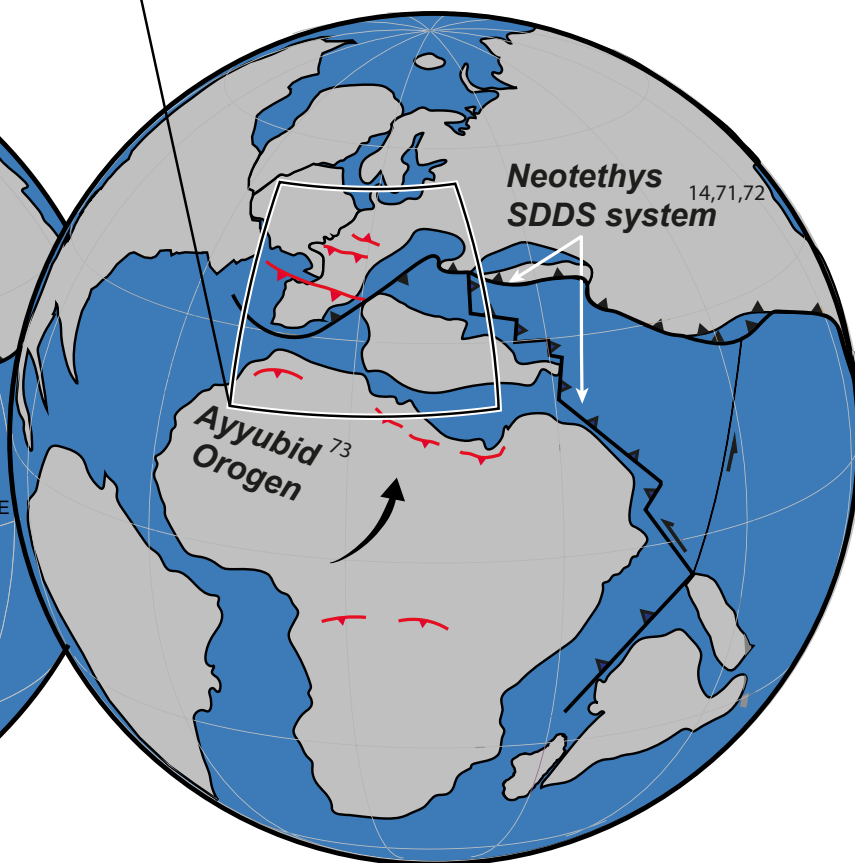


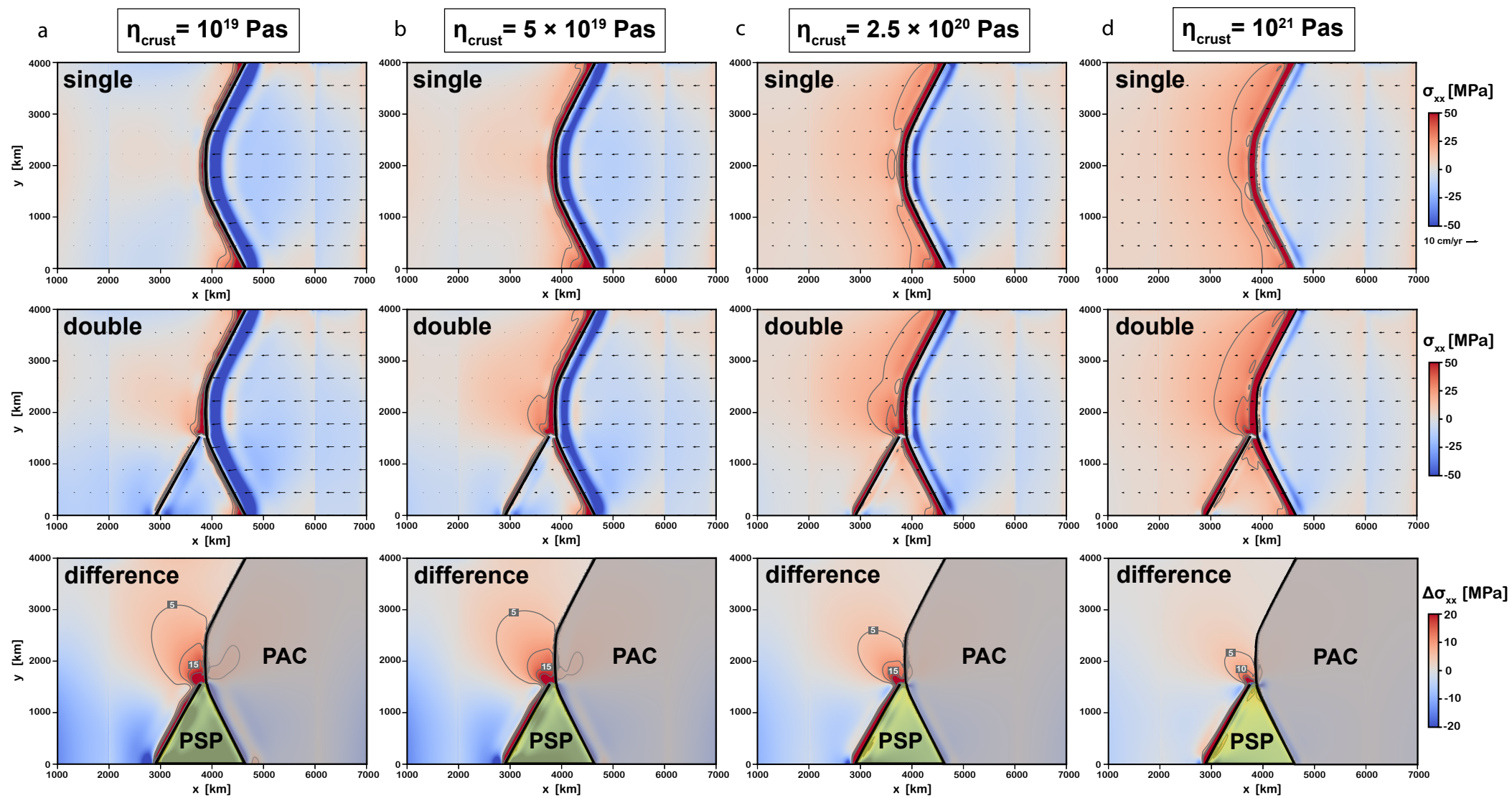
b

126 Myr ago¹⁴



92-70 Myr ago¹⁴





a

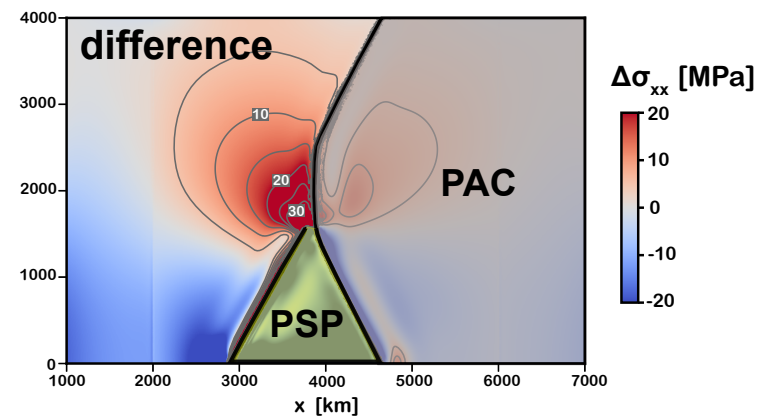
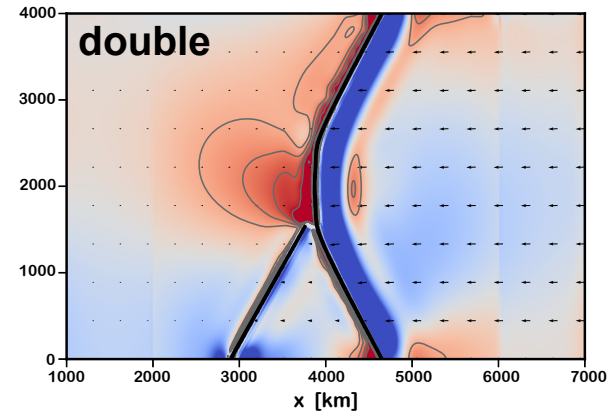
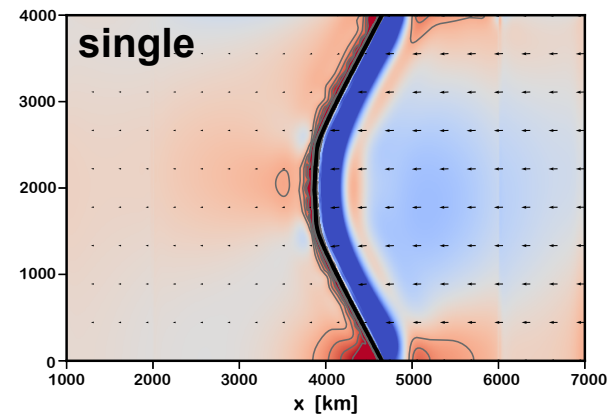
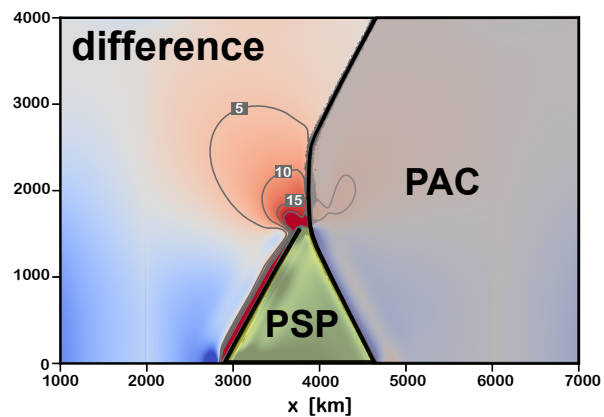
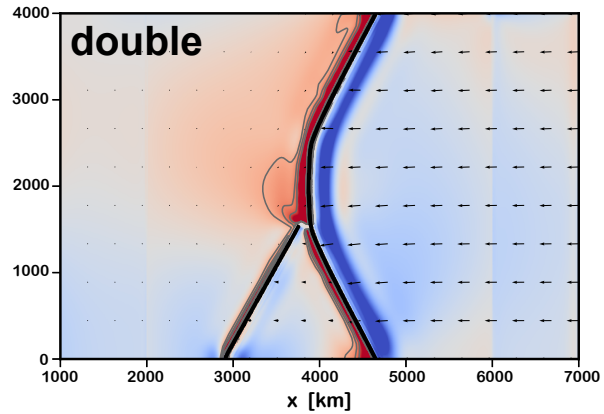
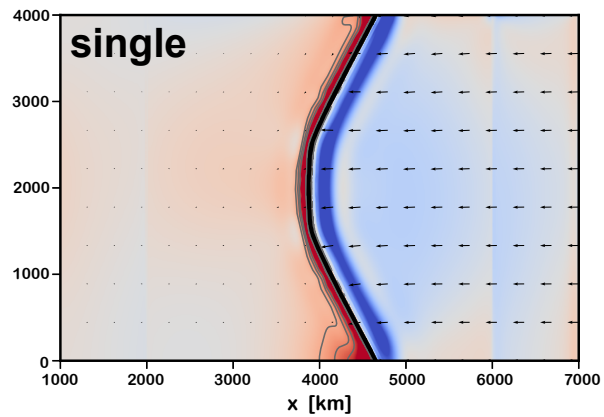
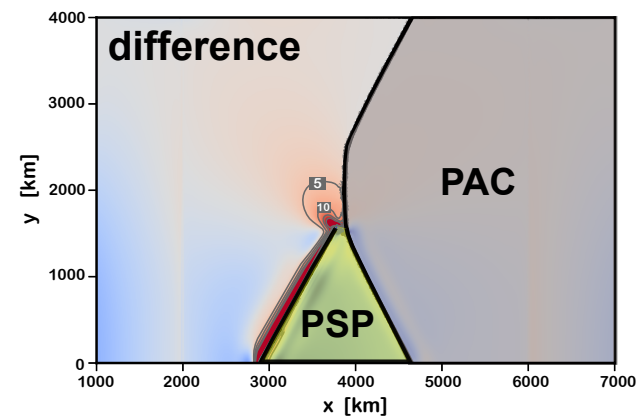
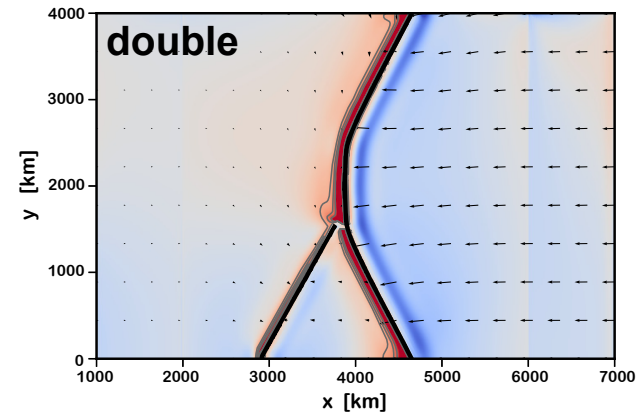
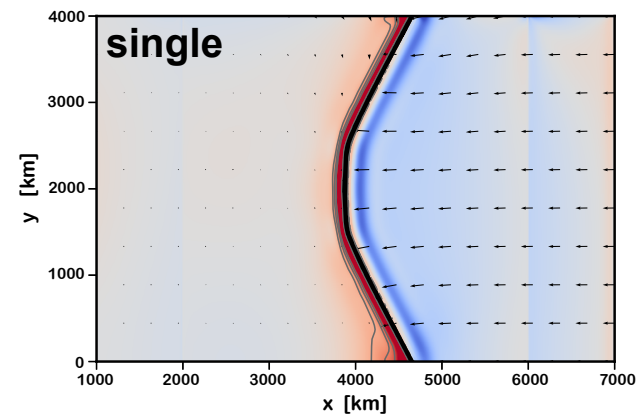
$$\eta_{\text{lith.}}/\eta_{\text{mantle}} = 125$$

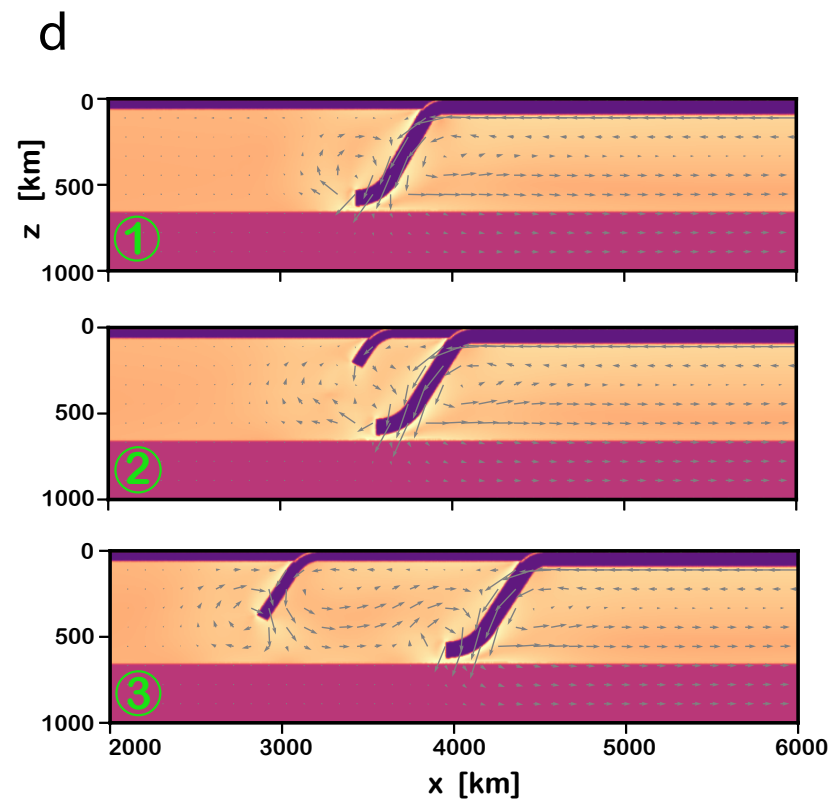
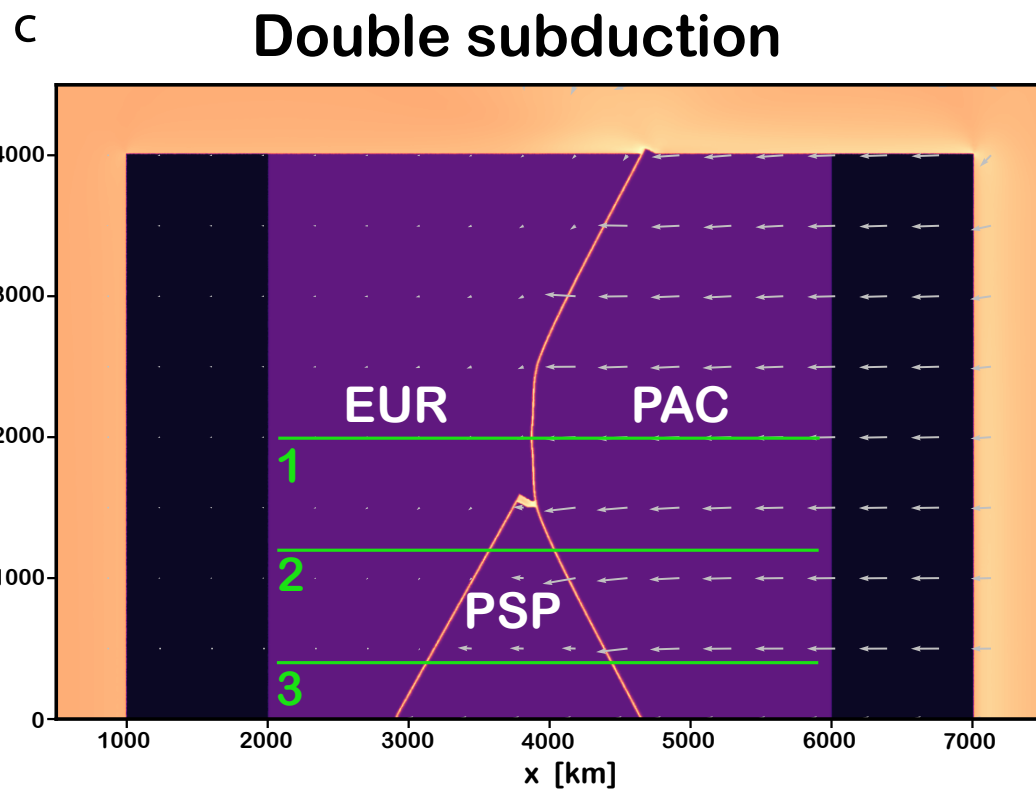
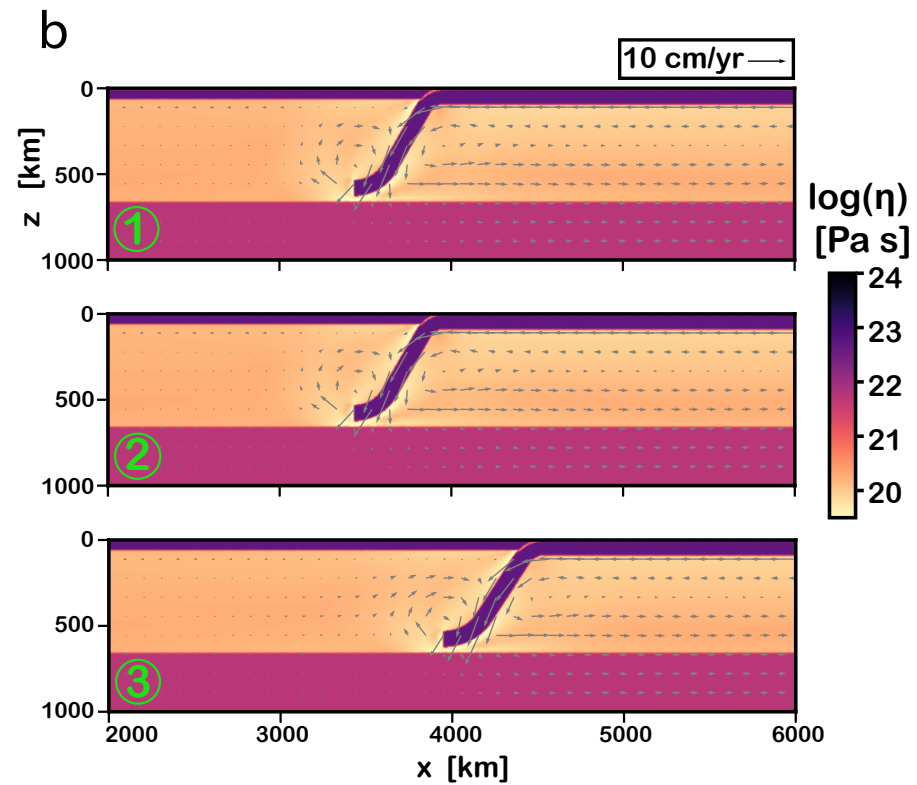
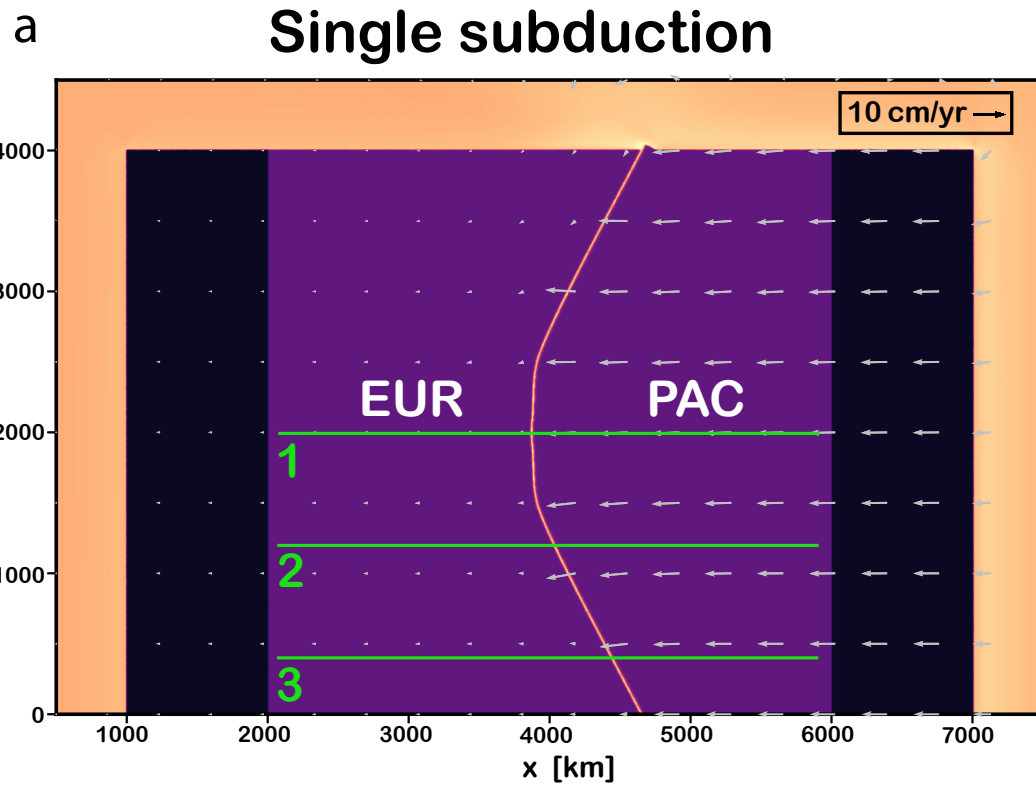
b

$$\eta_{\text{lith.}}/\eta_{\text{mantle}} = 500$$

c

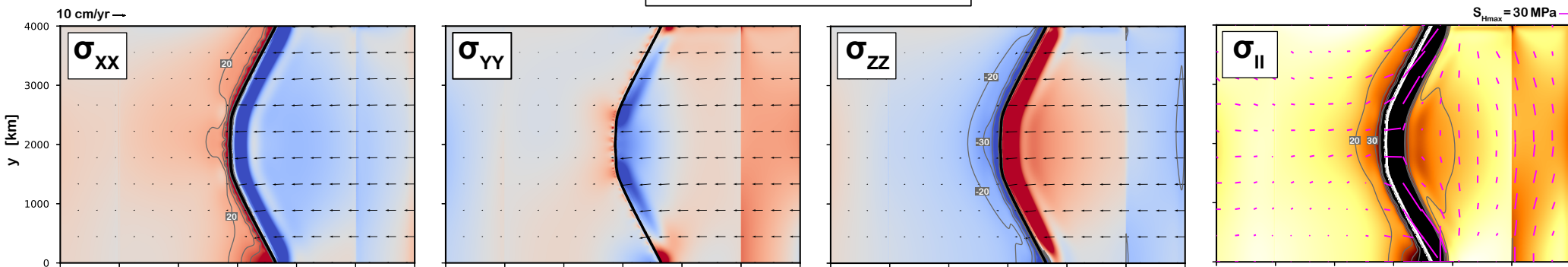
$$\eta_{\text{lith.}}/\eta_{\text{mantle}} = 2000$$





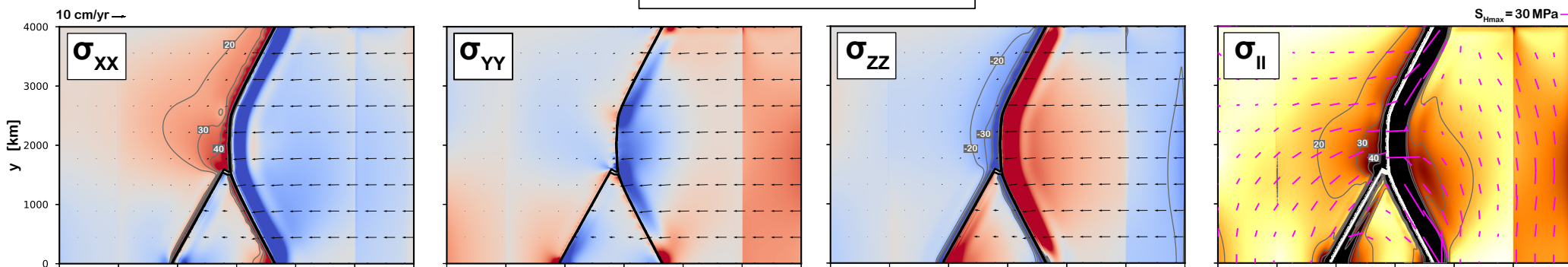
a

Single subduction



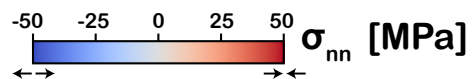
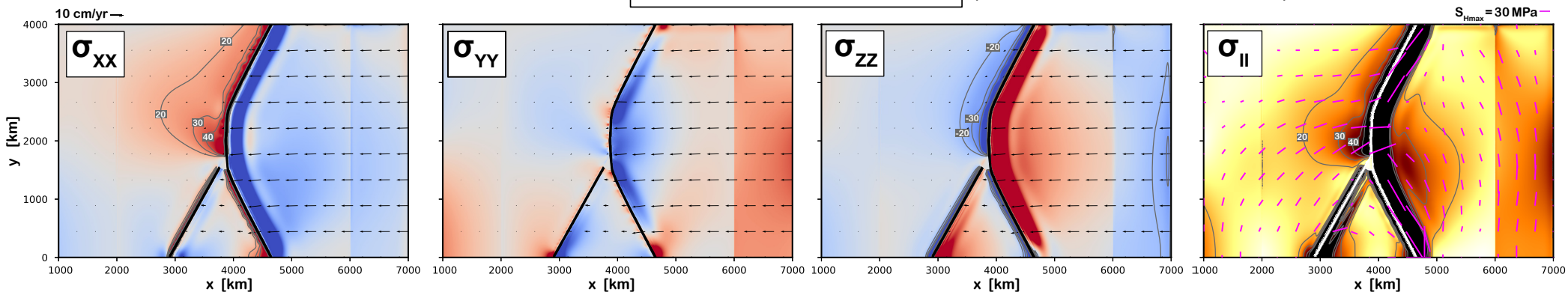
b

Double subduction

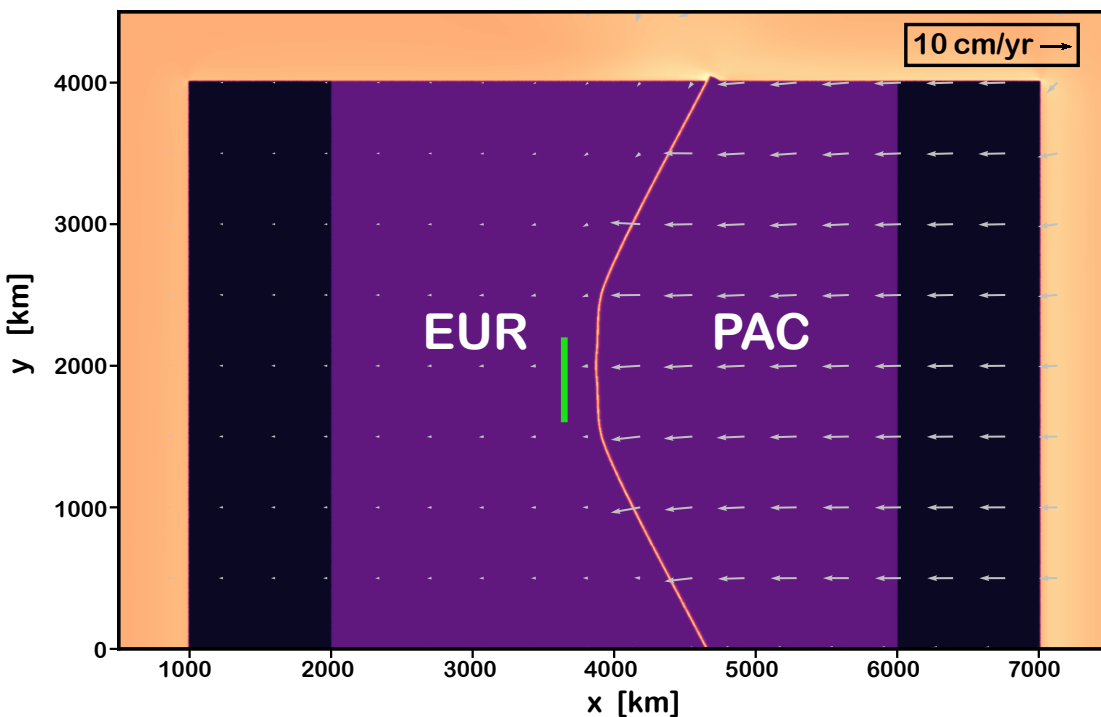


c

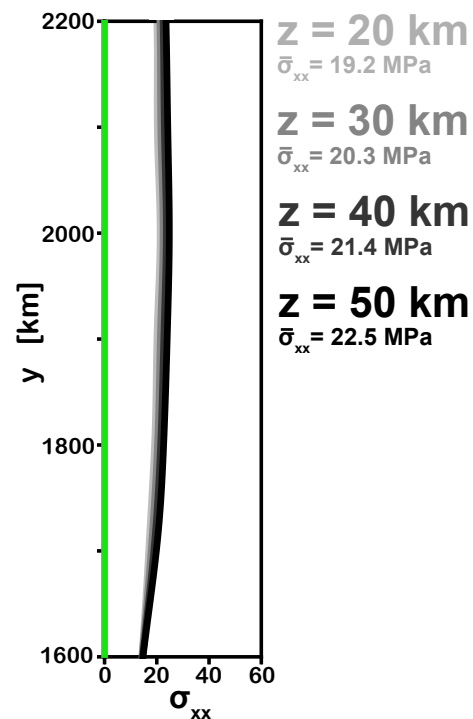
Double subduction (alternative weak zone shape)



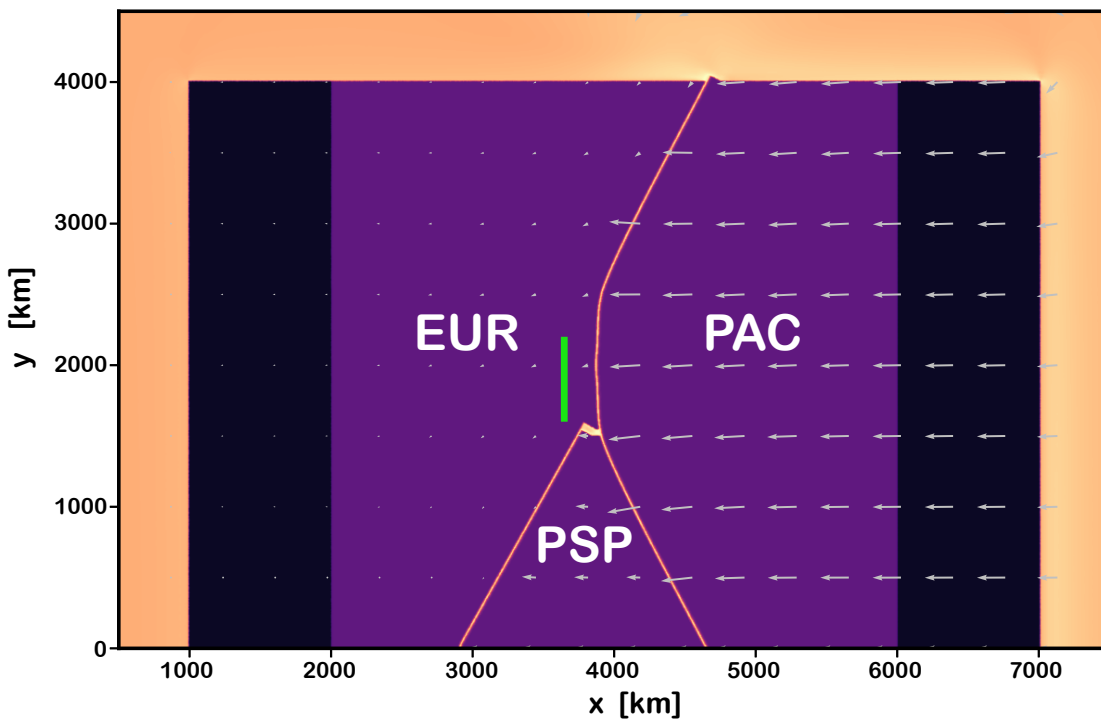
a Single subduction



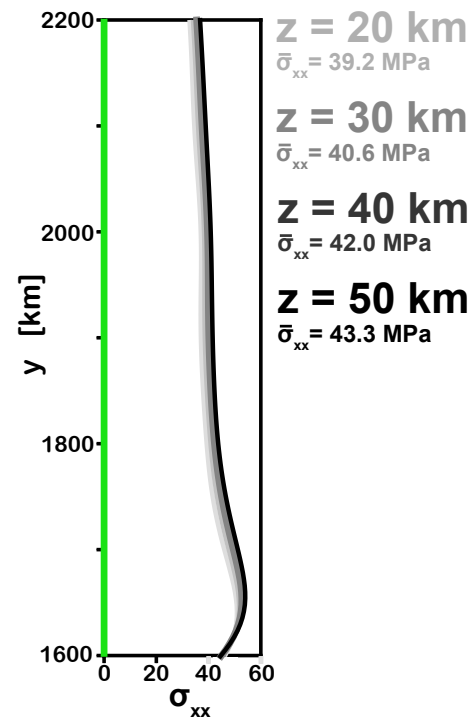
b



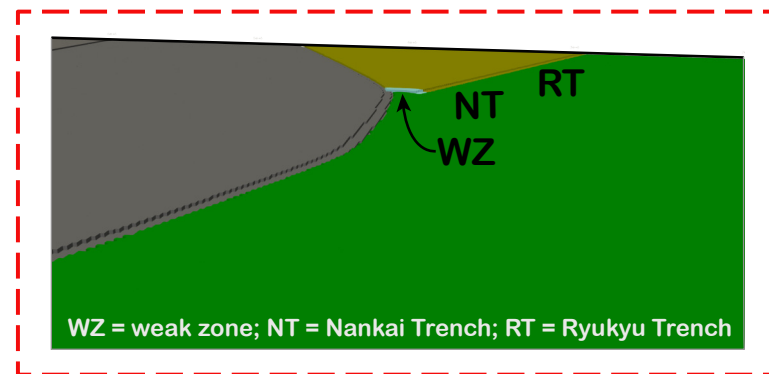
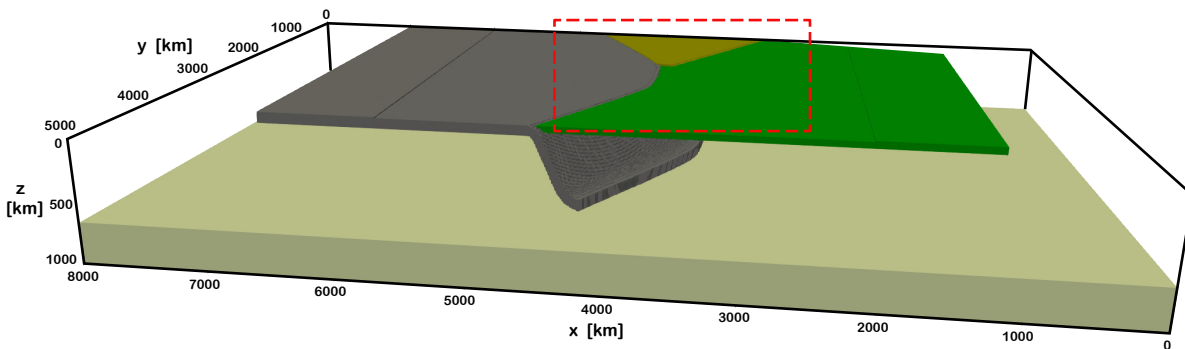
c Double subduction



d



a : All tectonic plates



b : Without overriding Eurasia

